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import numpy as np
import matplotlib.pyplot as plt

# 1. Define the convex curve (e.g., a simple parabola)
x = np.linspace(0.5, 5, 100)
y = 0.5 * x**2 + 1

# 2. Define Point P in the middle of the curve
x_p = 3.0
y_p = 0.5 * x_p**2 + 1

# Define point Q on the curve to the lef of P
x_q = 2.5
y_q = 0.5 * x_q**2 + 1

# 3. Calculate the tangent line at Point P
# The derivative of y = 0.5*x^2 + 1 is dy/dx = x
m_p = x_p  # slope at Point P
# Equation of tangent line: y_tan = m*(x - x_p) + y_p
x_tan_p = np.linspace(0, 4.5, 100)
y_tan_p = m_p * (x_tan_p - x_p) + y_p

# Calculate the tangent point at point q
m_q=x_q  # slope at Point Q
x_tan_q = np.linspace(0, 4.5, 100)
y_tan_q = m_q * (x_tan_q - x_q) + y_q


# 4. Calculate the y-intercepts (b)
b_p = m_p * (0 - x_p) + y_p
b_q = m_q * (0 - x_q) + y_q

# 5. Create the plot
fig, ax = plt.subplots(figsize=(6, 5))

# Plot the convex curve (blue line)
ax.plot(x, y, 'b-', linewidth=2, label='$y = f(x)$')

# Plot the tangent line (dashed line)
ax.plot(x_tan_p, y_tan_p, 'k--', linewidth=1.5)
ax.plot(x_tan_q, y_tan_q, 'k--', linewidth=1.5)

# Plot Point P (black circle)
ax.plot(x_p, y_p, 'ko', markersize=6)
ax.text(x_p + 0.1, y_p - 0.5, 'P', fontsize=12,
 ↪  fontweight='bold')
# Plot point Q (black circle)
ax.plot(x_q, y_q, 'ko', markersize=6)
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ax.text(x_q + 0.1, y_q - 0.5, 'Q', fontsize=12,
 ↪  fontweight='bold')

# Plot the y-intercept (b)
ax.plot(0, b_p, 'ko', markersize=6)
ax.text(-1.0, b_p, 'b_P', fontsize=12, fontweight='bold')
ax.plot(0, b_q, 'ko', markersize=6)
ax.text(-1.0, b_q, 'b_Q', fontsize=12, fontweight='bold')

# Labeling and formatting
ax.text(3, 10, '$y = f(x)$', color='blue', fontsize=12)
ax.text(1, -2, '$y = m*x + b$', color='black', fontsize=12)

ax.set_xlim(0, 5.5)
ax.set_ylim(-4, 14)
ax.axhline(0, color='black', linewidth=1) # x-axis
ax.axvline(0, color='black', linewidth=1) # y-axis
ax.set_xlabel('x', fontsize=12)
ax.set_ylabel('y', fontsize=12, rotation=0)

# Remove top and right spines for a cleaner look
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.show()
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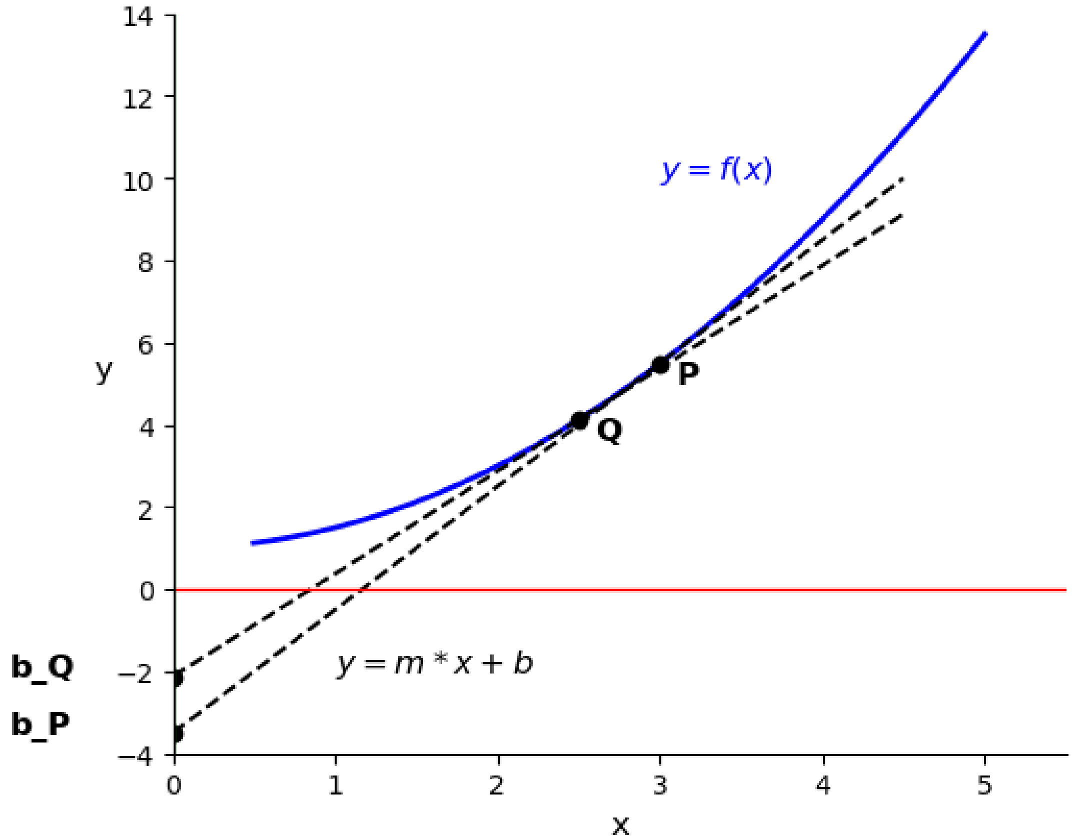
14
12
10
8
6
4
2
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−2
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y
b_Q
b_P
0
1
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3
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5
x
$y = f(x)$
P
Q
$y = m * x + b$